DRAFT: Spatiotemporal distributions of neustonic species in the genus *Physalia*

Dalila Destanović, *Samuel H. Church*, Namrata Ahuja, Maciej Mańko, Ramdhan Firdaus, Sarah Kingston, Adam Riggs, Ali Fox, Kylie Pitt, Amandine Schaeffer, Chris Simon, Inma Herrera, Mary Beth Decker, Casey W. Dunn

corresponding author: samuelhchurch@gmail.com, SHC

Abstract

The organisms at the ocean surface, termed neuston, experience unique environmental conditions at the interface of ocean and atmosphere. Recent research has shown this community to be diverse and dynamic, containing both cosmopolitan species as well as overlooked diversity, distributed by systems of currents and winds. Here we analyzed the spatiotemporal distribution of neustonic organisms in the genus *Physalia*, known as Portuguese man-o'-war or bluebottles, recently revised to include at least five species with distinct but overlapping ranges. We first classified 3,599 images of *Physalia* uploaded to the participatory science network iNaturalist, and used these as training data for two approaches: first, to fine-tune a machine learning model, and second, as a ground-truth dataset for a group of community labelers to classify the remaining images. Based on the combined annotation of more than 20,000 images, we describe species distributions over space and time at regions of range overlap (e.g. Eastern Australia, South Africa), and identify different seasonal dynamics leading to species-specific peaks in stranding each year. We characterize several regions that experience notable seasonal gradients (e.g. for *P. physalis* in the Western Atlantic), and further show these gradients are repeated across ocean basins (e.g., *P. megalista* peak in Oct.-Dec. at 30'S in the Pacific, Indian, and Atlantic oceans). For three species overlapping in the South Pacific, we unite these data with mitochondrial genomic data derived from open water collected specimens and use these to show that species inhabit distinct open ocean ranges that are repeated across sampling years. We additionally include new genomic data from *P. minuta* in the South Atlantic, extending the previously known range of this species. Finally, we present a morphological comparison of live and preserved specimens of the five *Physalia* species, distinguish aspects of their colony-level

organization, and designate neotype material for three species. The combination of thousands of labeled photographs, 168 publicly available mitochondrial genomes, and diagnostic morphological characters will facilitate future identification and allow for continued monitoring and research of this critical neustonic system.

Introduction

Open ocean environments have recently been shown to host significantly more diversity and population structure than previously thought (Norris 2000; Peijnenburg and Goetze 2013; Bowen et al. 2016; Johnson et al. 2022). These results challenge assumptions about a well-mixed ocean and suggest instead that biological and environmental processes play a dynamic role in maintaining boundaries between populations. Many of these results have been framed in terms of biogeography, describing genetic distances across spatial gradients to test, among other things, the role of isolation by distance (Casteleyn et al. 2010; Mills, Lunt, and Gómez 2007). But spatial divisions can tell only one side of the story (Goetze 2011, 2003; Unal and Bucklin 2010); temporal divisions, such as seasonal patterns, are also critical in shaping interactions. Cyclic fluctuations in ocean currents, winds, temperatures, and ecosystems play a significant role in defining species ranges and potential points of contact (Gaylord and Gaines 2000; White et al. 2010; Sunday, Bates, and Dulvy 2012). These processes are particularly relevant to the ocean-surface ecosystem, termed the neuston, at the air-water interface. This unique assemblage represents one of the most enigmatic and understudied ecosystems in the ocean, and one that is uniquely imperiled by pollutants that accumulate at the ocean surface (Helm 2021). Exploring spatiotemporal dynamics in neustonic communities presents an ideal opportunity to dissect the relationship between oceanographic processes and biodiversity across the range.

To test the role of temporal structuring of ocean communities, we require high-resolution data over time and space of species abundances. For many organisms, such data can be difficult to gather, given the massive distances to be monitored and the remote nature of open ocean environments. But for select organisms, participatory or citizen science can give us large sample sizes through crowd-sourced observations. This is true for *Physalia*, a genus of open ocean cnidarians (Hydrozoa: Siphonophora) that sail the ocean surface and regularly wash ashore. On the participatory science website iNaturalist, there are more than 20,000 research-grade photographs of *Physalia* from dozens of countries and across all seasons. These observations are often vetted by multiple iNaturalist identifiers, such that the data are largely reliable and associated with exact time and place. Analyzing patterns in the seasonal abundance of *Phyalia* across the range presents an opportunity to unpack seasonal structure in biodiversity at the ocean surface.

In our previous study of *Physalia* diversity (Church et al. 2025), we used population genomic data to show that there are at least four species in the open ocean, each with a unique oceanic distribution and identifiable morphology. We further showed that within each species, there are patterns of regionally localized subpopulations with unique genomic signatures, as well as evidence for long-distance migration events between subpopulations. These species overlap in parts of their distribution, but it is unclear how species boundaries have been maintained. One open question is to what extent these species co-occur within the same season, or if they present distinct seasonal distributions that might reflect distinct source regions, or differential movement in response to ocean currents and temperature. Answering these questions is essential as *Physalia* represent a significant component of the neustonic ecosystem, serving as a key predator and fish-specialist(Damian-Serrano et al. 2022; Purcell 1984; Nunes et al. 2025). They also have potentially significant impacts on human health and tourism, given the medical relevance of their sting when encountered in near-shore environments (Hewitt et al. 2026). Understanding seasonal fluctuations in the abundance of individual species, therefore, has direct implications for how to predict and manage human-*Physalia* interactions.

Here we use a combination of machine learning and participatory science to identify and describe patterns in the abundance of *Physalia* species around the world. We test for both seasonal gradients in beaching events and investigate regions of species biogeographic overlap to determine whether species exhibit differential peaks in abundance. For one such region, the Southwest Pacific, we incorporate new open ocean genetic sampling to connect patterns in strandings to open ocean distributions. With new beaching samples and iNaturalist observations from the South Atlantic, Indian, and Southeast Pacific, we expand the known range of *P. minuta* previously only found in the Southwest Pacific Ocean. Finally, to aid the future monitoring of *Physalia* across the globe, we present novel morphological characterizations of colony organization and architecture, along with newly designated type specimens for three species. We combine these descriptions with thousands of labeled photos, a publicly available trained model that is 83.8% accurate, and 168 annotated mitochondrial genomes for molecular barcoding. Taken together, our results present a comprehensive community toolkit to identify *Physalia* around the world, which will advance our understanding and monitoring of *Physalia* stranding events across our shared ocean.

Methods

Participatory science images

iNaturalist.org was accessed for participatory science observations on several occasions across the lifetime of the project. An initial dataset of 20,007 images was downloaded on June 4, 2025, and used for training. A final dataset of 20,734 images was downloaded on August 14, 2025. In addition, datasets on other commonly observed beach species (hermit crabs, echinoderms) were downloaded on August 22, 2025. All datasets analyzed here are included at https://github.com/shchurch/Physalia_spatiotemporal_patterns. Images not associated with a precise location or timestamp data were excluded (31 images from the final *Physalia* dataset).

A subset of ~4,000 was examined manually and preliminarily assigned to species based on the characters described in Church *et al.*, 2025 (Church et al. 2025). These were preferentially selected to include key regions of species overlap (New Zealand, Brazil, etc.) as well as regions with only one previously known species (Eastern North America, Hawai'i), to create a balanced and rich training dataset including multiple sizes, angles, and preservation conditions. From this subset, 2,706 images could reliably be identified to species and were selected as the initial training dataset.

Image classification via semisupervised learning

The initial training dataset was used to fine-tune a pretrained model, google/vit-base-patch16-224-in21k (a Vision Transformer, ViT), using Python with PyTorch and the HuggingFace Transformers library. Model tuning was performed for 15 epochs, after which the model was redeployed to classify the remaining 17,163 images. High-confidence (>90%) images were examined for their distribution over geographic range, and images falling outside previously understood ranges for each species were manually examined. Labels were either accepted, reassigned, or images were marked as likely unidentifiable given current morphological understanding (for example, juvenile specimens with a globular float and few colony bodies were deemed indistinguishable). Range boundaries for inspecting outliers were originally described as follows: *P. physalis*, records occurring outside the Atlantic basin (Indian and Pacific Oceans, longitude $> 30°$E; northeastern Pacific, longitude $< -100°$W, latitude $> 0°$N; southeastern Pacific, longitude $< -60°$W, latitude $< 0°$S); *P. megalista*, records north of 15°S; *P. utriculus*, records in the North Atlantic (longitude $-100°$ to 20°E, latitude 15°–70°N) and New Zealand region (latitude $-49°$ to $-33°$S, longitude 165°–180°E); *P. minuta*, records north of 15°S or west of 110°E.. Confirmed new labels were then added to the original training dataset, and the process was repeated for two generations. This process resulted in a final training dataset of 3,599 manually confirmed images and 1,041 records flagged as unidentifiable.

A final image classification model was fine-tuned on the final training set, using an ensemble of three google/vit-base-patch16-224-in21k models with temperature scaling calibration and focal loss (= 2.0), for 10 epochs, using a 70/15/15 split for training, testing, and validating the training data. Final model accuracy was evaluated using a confusion matrix (Figure 1 C-D); overall accuracy on the held-out set was 83.8%. This model was deployed on the remaining images, of which 7,566 were assigned labels above a confidence threshold of >65% and were incorporated into the spatiotemporal dataset.

Community labeling images

The final training dataset of 3,599 manually confirmed images was provided to the iNaturalist community, who initiated a project to relabel the remaining images around the globe. Authors S.H.C. and C.W.D. assisted the community in this work, responding to specific questions and comments on iNaturalist as tagged. iNaturalist community members determined the best course of action for relabeling, opting to use atlases for initial identifications based on previously understood ranges, and evaluating images in areas with known overlap. By August 14, 2025, 13,979 images had been labeled to species, at which point labels were downloaded and incorporated into the spatiotemporal dataset.

Final labels for the spatiotemporal dataset were determined by combining the training, classified, and community label datasets, giving priority to the trained labels, and then the community labels, if conflicts arose. Labels for images marked as unidentifiable during initial training were removed as potentially unreliable. A final manual inspection of geographic outliers (n=2,260) was performed to confirm the identity of any images that would extend the known range of species. After finding genetic evidence that the range of *P. minuta* extends to the South Atlantic (see below), authors S.H.C. and D.D additionally re-examined manually corrected samples (679), as well as all observations from the South Atlantic Ocean (729) and Indian Ocean (1,158).

Neither iNaturalist nor machine learning-defined labels could differentiate between *P. utriculus* and *P. mikazuki* (Yongstar et al. 2025, 2026), as there is currently not enough information on the differences in their morphology for high-confidence species identification (see discussion below).

Seasonal abundance characterization

To visualize potential seasonal patterns, we combined classified observations across years and visualized abundance in three ways: [1] spatially, across four three-month epochs of the year; [2] daily, counting observations within defined regions; and [3] calculating the median calendar day of observations across a grid to analyze spatiotemporal gradients. To account for potential confounding effects of iNaturalist user effort, we normalized to observations of abundant and commonly reported beach species (hermit crabs and echinoderms). Normalization followed two approaches: a raw ratio of daily *Physalia* observations to proxy beach species, and a normalization using a generalized additive model (GAM) with a cyclic cubic spline fit to proxy species daily counts that accounts for seasonal trends. Further, comparing differences in species abundances in the same region provides an internal control for user effort, as iNaturalist observations were originally recorded agnostic to species.

Genetic and phylogenetic analyses

Sample collection

Specimens were collected in the Southwest Pacific and South Atlantic. The Southwest Pacific collections happened during two cruises aboard the Sea Education Association's SSV Robert C Seamans. The first cruise took place from March 26th to May 3rd, 2024, from Lyttelton, Aotearoa, New Zealand to Pepeete, Tahiti, French Polynesia. The second `cruise took place from April 3rd to May 16th 2025 from Lyttelton, Aotearoa, New Zealand to` Moorea, French Polynesia. Physalia was collected twice a day with 335 µm mesh neuston tows at a speed of 2 knots for 0.5 hour at 10:30 am and 10:00 pm, weather permitting. Specimens were separated from the rest of the zooplankton sample, measured individually, phenotyped for handedness, and photographed. Tentacle tissue was subsampled and preserved in a DNA shield at -20°C (Zymo Research, Orange, CA, USA), while whole specimens were preserved in 100% ethanol. Additional beached specimens from this region were collected in Breaker Bay, Foxton Beach, and Lyall Bay. Also, we received one specimen from the Western Australian Museum collected in the Eastern Indian Ocean. The South Atlantic specimens were collected at Sandy Beach, Saint Helena, during two periods: September 24th, 2025, and January 11th to 20th, 2026. Specimens were stored in 100% ethanol and, if in good shape, in formalin for morphological purposes, and photographed. Information on sequenced specimens can be found in Supp. Table 1 (`Supp_Table_specimens.tsv`).

DNA extractions

Samples were stored in 100% ethanol and DNA Shield (Zymo). The same procedure was used for ethanol (10 samples) and DNA Shield-stored samples(10 samples). A piece of a tentacle was used as input for all of the samples, except for one degraded sample from Saint Helena, where we used a piece of float (YPM-IZ-116019). DNA was extracted using the EZNA Mollusk Kit following the manufacturer's instructions, with an overnight digestion step at 60°C. Yale Center for Genome Analysis processed the genomic DNA and prepared libraries for Illumina NovaSeq paired-end 150bp whole-genome sequencing (WGS) (10x coverage).

Read quality control and mitochondrial genomes assembly

For subsequent analyses, we used previously published *Physalia* WGS libraries (n=151), and added the new Southwest Pacific and South Atlantic samples. The raw reads were checked with a quality control pipeline [put the pipeline on GitHub]. In this pipeline, we subset reads to 10 million, trim them with Trimmomatic (default settings) (Bolger, Lohse, and Usadel 2014), and run them through FastQC, Kraken2 (standard contaminant database)(Wood, Lu, and Langmead 2019) and sharkmer v1.0.1. (Dunn and Church 2026) (metazoa, cnidaria, and bacteria PCR primers). The reads were also mapped with BWA (Li 2013) to a human genome [GRCh38, GCA_000001405.15], to check for human contamination, and were used to assemble and annotate mitochondrial genomes using GetOrganelle (custom seed of siphonophore genomes, -R 10, –reduce-reads-for-coverage inf, –max-reads inf) (Jin et al. 2020), MITOS2 (–orih 0, –oril 0, –trna 0, –intron 0, –linear, -c 4) (Donath et al. 2019) and tRNAscan-SE (-O, -g gcode.invmito) (Chan et al. 2021).

Phylogenetic analyses

Phylogenetic trees were built using the *cytochrome oxidase I* (*COI*) gene and mitochondrial genomes. All samples above were used, and the outgroup is two species of *Rhizophysa.* For the *COI* tree, we used GQ120041, AY937377, GQ120040, and GQ120038 as outgroup sequences. COI sequences and mitochondrial genomes were independently aligned using MAFFT (Katoh and Standley 2013), and then the alignment was used for the generation of the phylogenetic tree using IQTree (bootstrap 1000) (Nguyen et al. 2015).

Morphological description

Physalia specimens are often encountered on the beach after washing ashore, where their anatomy may be compromised by physical damage or decomposition, depending on stranding conditions. We sought to describe these animals from live specimens to better understand differences in the natural morphology and posture of *Physalia* species as they appear floating in water. We collected and photographed *P. megalista* and *P. minuta* specimens from the North Island, New Zealand (May 2024), *P. physalis* from Gran Canaria, Canary Islands (March 2025), and *P. utriculus* from the Big Island, Hawai'i (April 2025) and observed macromorphological differences in colony-level features. We supplemented these observations with examination of fixed specimens and designated neotype material for the three species for which no type material currently exists. Additionally, to advance our understanding of *P. utriculus* and *P. mikazuki* in the Indian Ocean, we observed and photographed specimens from [**TODO-1: Indonesian collection locality, #2**], Indonesia.

Results

Identifying *Physalia* species in images

Our analysis of a dataset of global observations of *Physalia*, using a hybrid semisupervised machine learning approach and community labeling approach, resulted in 16,453 of 20,734 images identified to species level (Figure 1 A). This approach first employed manual labeling of 2,706 images and iterative fine-tuning of a classification model to generate a training dataset of 3,599 images (1,096 *P. physalis*, 1,293 *P. utriculus*, 1,032 *P. Megalista*, 178 *P. minuta*). These were used to tune a machine learning classification model and also provided as a ground-truth reference to community labelers on iNaturalist. The model resulted in 7,566 high-confidence classifications (threshold >65%, selected to balance assignments with accuracy), and the community made 13,979 assignments. This difference is likely due to the use of `atlases` (expected range maps) on iNaturalist to establish preliminary species IDs, which results in automatically assigned images that are then verified by the community of labelers. From the final dataset, 2,261 images that represented outliers from the known range of the species were manually confirmed by the authors, resulting in 794 corrections.

Comparing labeled images between the model and community labeling, we observed strong consistency, with 4,899 out of 5,022 joint observations being shared (97.6%; Figure 1 B). Model validation showed an accuracy >80% for all species even without filtering on confidence, as calculated using 15% of manually labeled images that were held out during training (Figure 1 C-D). Given the strong variation in image quality and composition, preservation of beached colonies, and the complex morphology of these colonial animals, we estimate this may be near the maximum possible accuracy. Visualizations of model attention (representing the pixels most highly weighted in making an assignment) show that the model is largely focused on the float and colony zones (Figure 1 E-F).

Seasonal dynamics in regions of range overlap

Our previous study describing species distributions had identified several regions of range overlap between *Physalia* species, raising questions about potential ecological overlap and mechanisms maintaining species barriers. To investigate the potential role of differential seasonal partitioning, we used the classified images to assess seasonal abundances in four regions: the Southwest Pacific (Figure 2 A-B), Southern Africa (Figure 2 C-D), the Caribbean and North America (Figure 3 A-B), and the Southwest Atlantic (Figure 3 C-D).

Our results show striking differences in seasonality and fine-scale spatial distributions across species. In the Southwest Pacific (Figure 2 A-B), *P. minuta* and *P. utriculus* exhibit strong seasonal turnover, such that *P. minuta* is virtually unobserved in Southern Hemisphere winter months (no observations in Jul.-Aug.-Sep.), while *P. megalista* is present most of the year, with a peak in Sep. and Oct. [**TODO-28: subject switch, #5 — this sentence opens on *P. minuta* and *P. utriculus* but the contrasting clause discusses *P. megalista*. Panel B supports the contrast as written (both *minuta* and *utriculus* drop out in winter; only *megalista* is year-round), so the likely fix is to the opening pair rather than the contrast — but which pair was intended is an author call.**] We also find that in summer months (Jan.-Feb.-Mar.) *P. utriculus* is highly abundant in Eastern Australia but rare in New Zealand and Tasmania, and vice versa for *P. minuta*. Strong overlaps between species in both space and time include between *P. utriculus* and *P. megalista* in Eastern Australia in Oct.-Nov.-Dec., and *P. minuta* and *P. megalista* in New Zealand in the same months.

These patterns are consistent across ocean basins. In Southern Africa (Figure 2 C-D) *P. megalista* again shows a peak in Sep. and Oct., and is rare in Jan.-Feb.-Mar. *P. utriculus* and *P. minuta* are less abundant in Jul.-Aug.-Sep., consistent with observations from the Southwest Pacific. These trends are robust to seasonal changes in iNaturalist user effort, as measured using abundant beach species (hermit crabs, echinoderms) as a proxy for observer activity. Seasonal patterns persisted or were even stronger when effort was taken into account- see, for example, observations of *P. utriculus* in Southern Africa ([**TODO-2: effort-corrected**

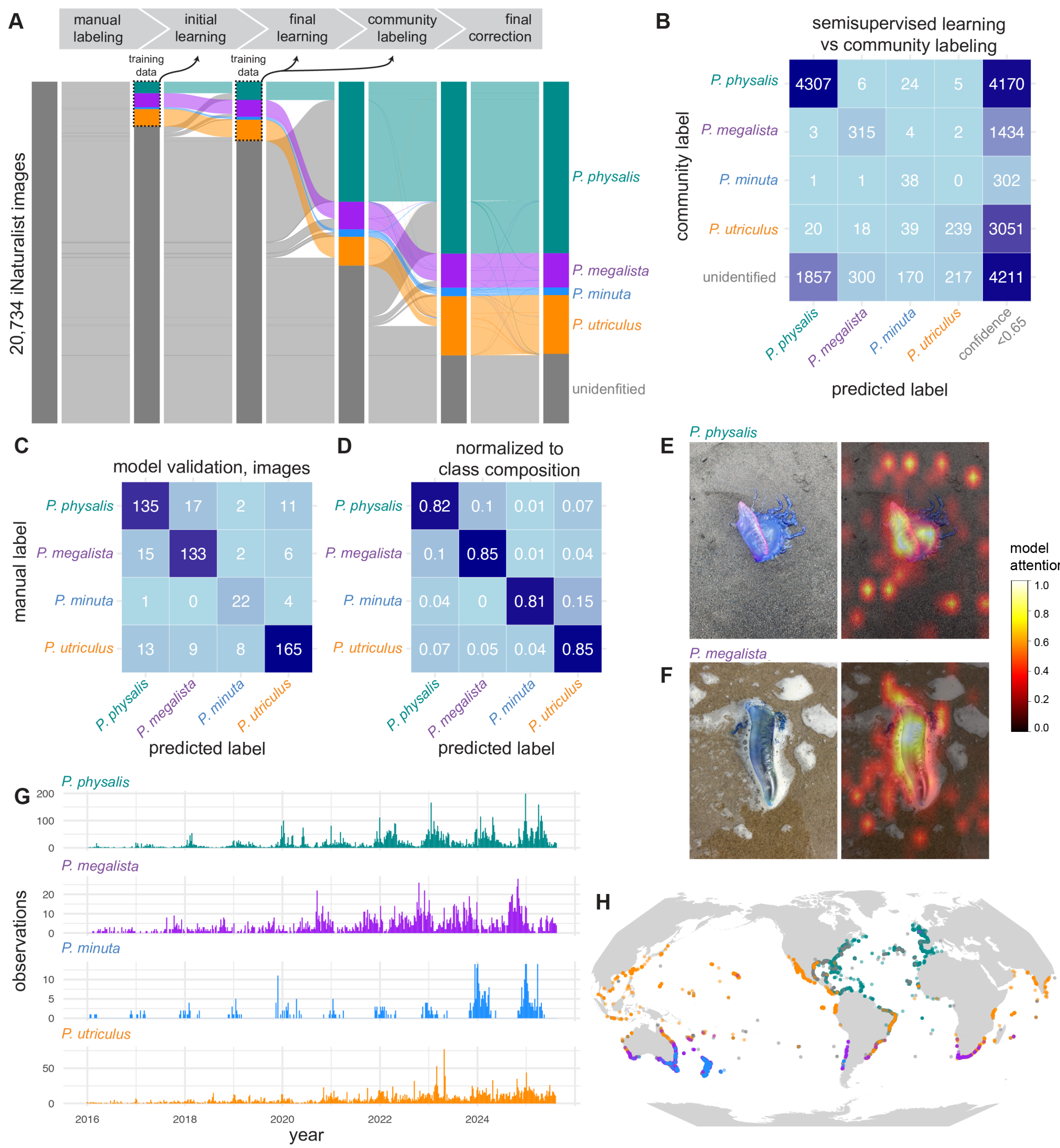

Figure 1: **Integrating semisupervised learning and community labeling to assign species to thousands of images.** A, 20,734 images from iNaturalist were processed for species assignment. 2,706 images were manually labeled, then used as training data for iterative classification and labeling steps, resulting in a final training set of 3,599 images. These were provided as training data for an image classifier as well as community scientists via iNaturalist. Final assignments were inspected for geographic and temporal outliers. B, Correspondence between high confidence (>0.65) predictions and community assignments. C-D, Confusion matrix comparing model predictions to manual labels, shown as (C) number of images and (D) proportion, normalized to manual labels per class. E-F, Visual attention of the trained classifier on sample assigned images for (E) *P. physalis* and (F) *P. megalista*. Image credit, [**TODO: image credit**]. G-H, Temporal (G) and spatial (H) distribution of the final labeled dataset of 16,453 images.

supp. figure, #23]): an anomalous day of high counts around the 116th day of the year is seen for both *P. utriculus* and other beach species, corresponding to a local holiday. When this discrepancy is accounted for, seasonal trends become more apparent. *P. minuta* had previously only been reported from New Zealand and Eastern Australia, but we recovered *P. minuta* in the South Atlantic (see below)These observations are corroborated by iNaturalist observations in Southern Africa, which were manually validated by the authors here. *P. minuta* is observed in the winter months, and *P. physalis* has been observed to wash ashore in Southern Africa in these same months, therefore, Oct-Nov-Dec represents a period of four-species overlap.

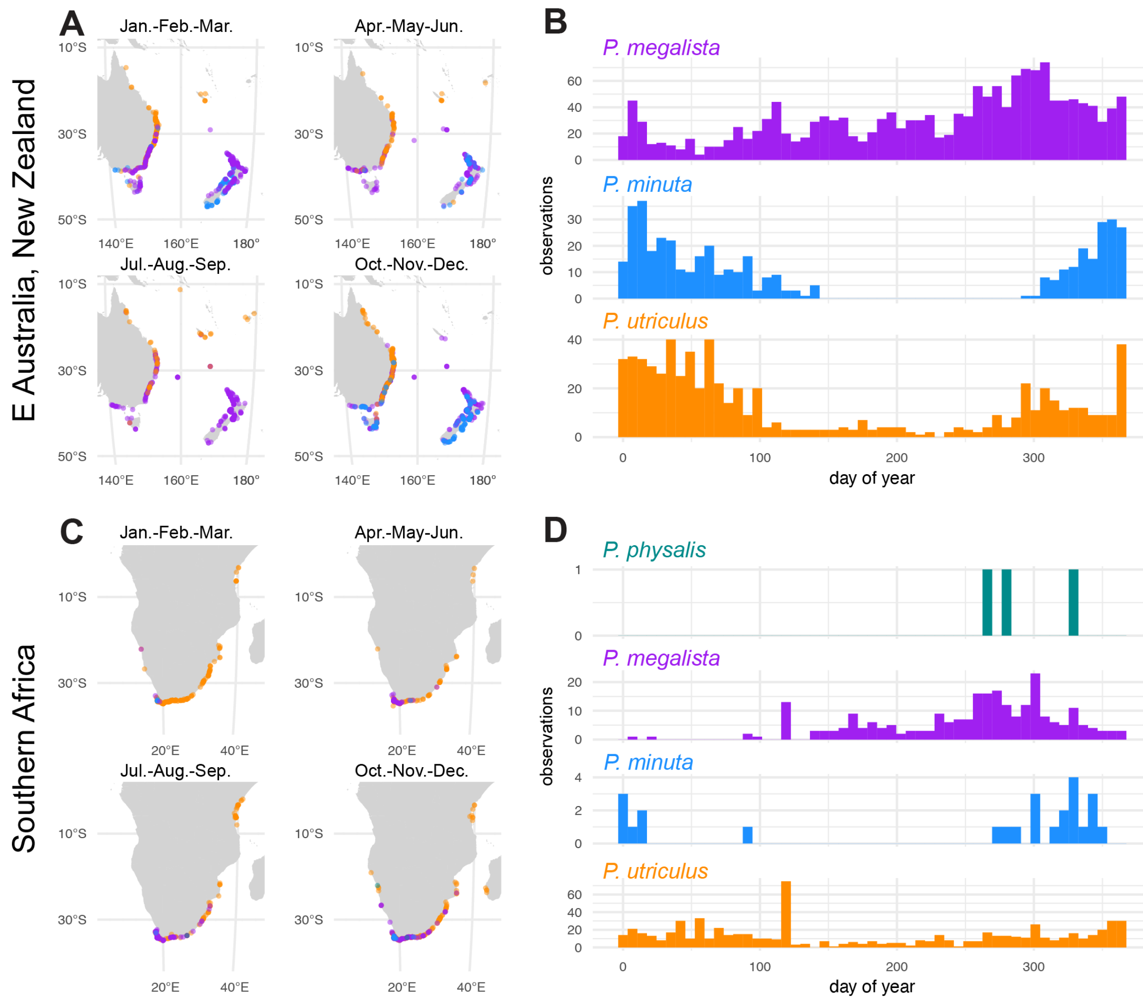

Figure 2: **Seasonal dynamics in the Southwest Pacific and Southern Africa.** (A,C) Distribution of high-confidence labeled images using a semisupervised ML approach in Eastern Australia and New Zealand (A) and Southern Africa (C). (B,D) Seasonal abundances, combined across years, for species in the regions shown.

In contrast to consistent signals across the Southern hemisphere, we observe distinct seasonal patterns for the same species between the Southern and Northern hemispheres. *P. physalis* in the Caribbean and Western North Atlantic (Figure 3 A-B) shows a strong depletion in abundance in Jul.-Aug.-Sep., but at more southern latitudes (Figure 3 C-D), the same depletion in abundance is seen earlier (Apr.-May.-Jun.). Here turnover between species is again apparent: in the Gulf of Mexico, *P. utriculus* is largely present when *P. physalis* is at its lowest abundance in Jul.-Aug.-Sep.. Seasons of particular overlap include Jan-Feb-Mar in the Western South Atlantic when *P. physalis*, *P. utriculus*, and *P. megalista* can potentially be found.

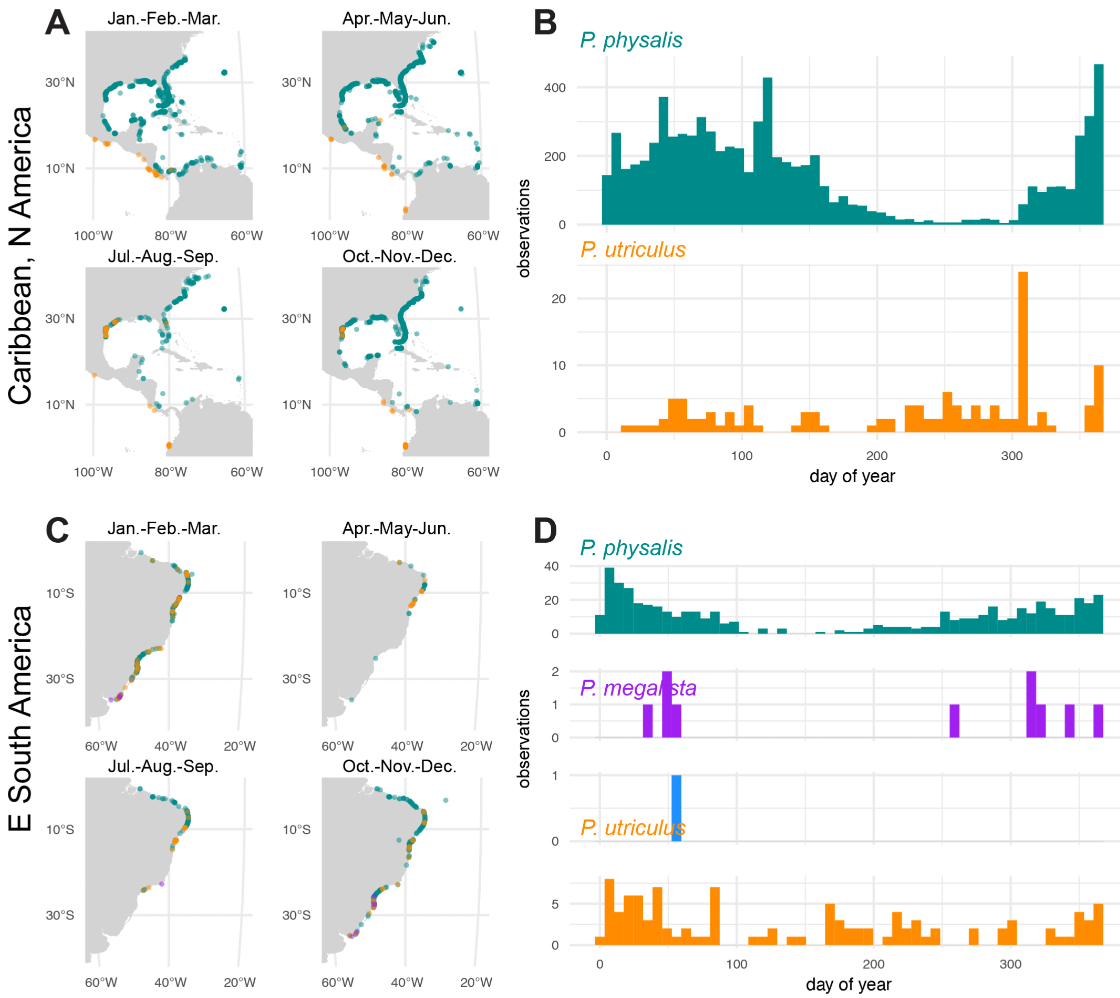

Figure 3: **Seasonal dynamics in the Western Atlantic.** (A,C) Distribution of high-confidence labeled images using a semisupervised ML approach in the Caribbean and North America (A) and Eastern South America (C). (B,D) Seasonal abundances, combined across years, for species in the regions shown.

Seasonal gradients across space and time

To examine seasonality across fine scales within a region, we visualized median day-of-year for observations across the grid for each species in the regions examined in the previous section (Figure 4).

In Southern Africa, we visualized the seasonal gradients of *P. megalista* and *P. utriculus* (Figure 4 C,D). The species' peak occurrences happen in different parts of the year: *P. megalista* peaks in Southern Hemisphere spring (Aug-Sep-Oct), and *P. utriculus* in the summer to fall (Nov to Apr), following a replacement trend. This suggests that even though these two species may overlap in space and time, they have distinct annual peaks in abundance that do not overlap. We also observe a latitudinal gradient in the distribution of *P. utriculus* on the eastern coast, ranging from up north in Tanzania (July) to the southern coast of South Africa (Jan-Apr), but as there are few observations from Mozambique, it is unclear whether the specimens observed in Tanzania and those in South Africa are a part of the same transportation system. In the Caribbean and North America (Figure 4 E,F), we observe a clear gradient for *P. physalis* from the Gulf of Mexico northward to the eastern coast of North America, ranging from late winter/early spring (Jan-Apr) to late summer (Jun-Sep) driven by the Gulf Stream, consistent with our previous study (Abedon et al. 2025). *P. utriculus* is observed in the Gulf of Mexico in the Northern Hemisphere fall (late Aug, Sep), a pattern similar to the replacement of *P. megalista* and *P. utriculus* in South Africa. We have also visualized trends for *P. megalista* and *P. utriculus* in the Southwest Pacific, as well as *P. physalis* and *P. utriculus* in the western South Atlantic ([**TODO-3: supp. figure, #23**]). *P. megalista* follows a latitudinal gradient from north (Sep) to south (Apr) on the eastern Australian coast in the Southwest Pacific ([**TODO-4: supp. figure panel, #23**]). The patterns observed for other species are more complex, and it is necessary to further study current patterns in the regions, as well as regions of high productivity, to interpret them.

Open ocean distributions in the Southwestern Pacific

In combination with our analysis of beaching observations, we collected open ocean samples from the Southwest Pacific over multiple years and performed whole-genome sequencing to confirm their identity. We combined these with sequenced specimens collected from beaches in the same region to investigate the offshore components of their seasonality, given that this region has three overlapping species with distinct spatiotemporal patterns of stranding.

Both *P. minuta* and *P. megalista* were collected from the open ocean east of New Zealand, with *P. minuta* consistently (over multiple years) recovered west of ~150 deg W longitude, and *P. megalista* observed further east, with the exception of a single *P. megalista* observation at 168.6 deg W in 2025. These observations are suggestive of distinct open ocean distributions within and without the South Pacific gyre (Figure 5).

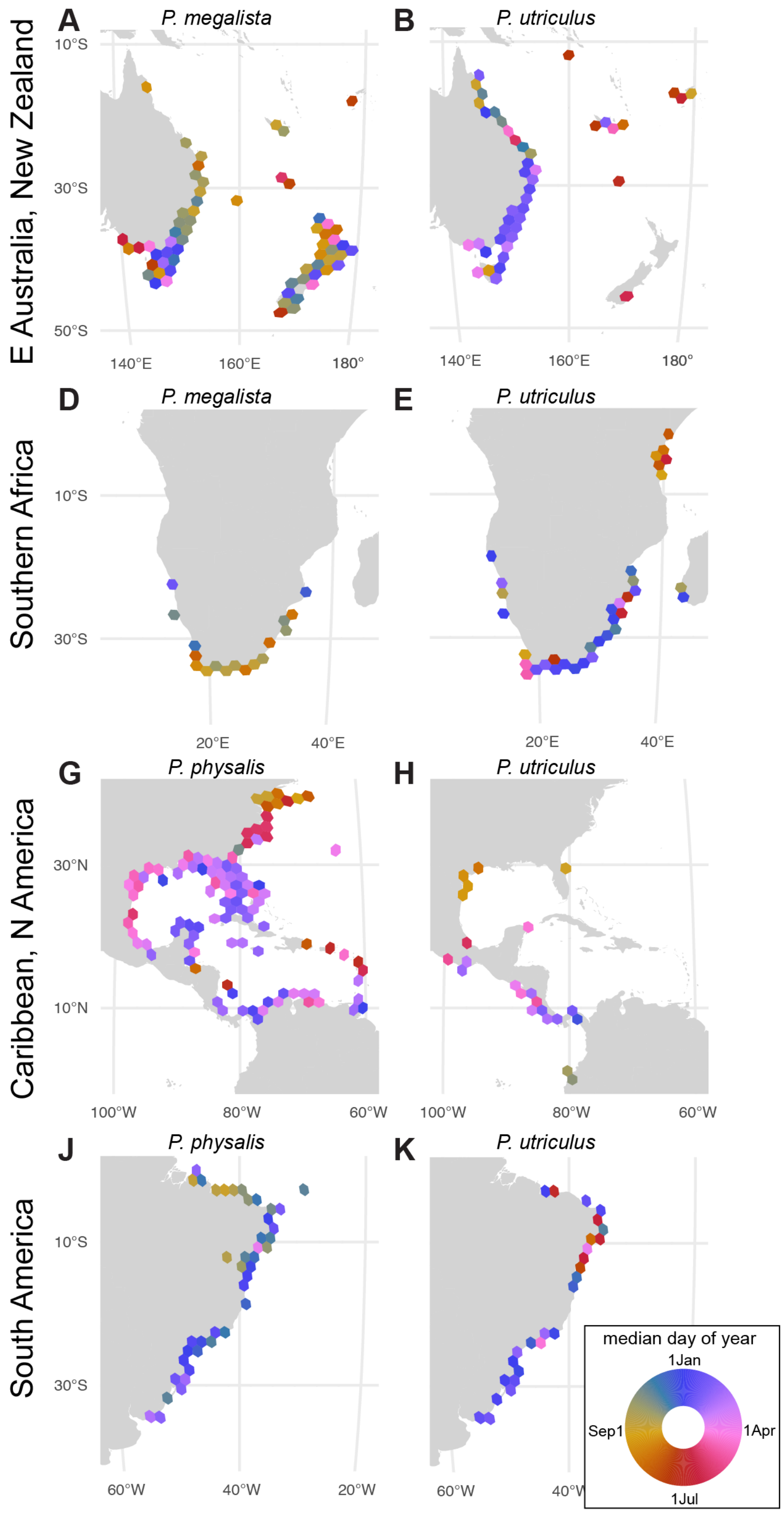

Figure 4: **Seasonal gradients across species and regions.** (A-K) Tiles indicate median day of year for species observations in that area. Strong seasonal gradients can be observed, e.g., in *P. physalis* northward up the W. Atlantic (A).

Open-ocean transects were sampled at the same time of the year (April-May), therefore giving insight into interannual signals, but cannot provide data on other time periods. Collections at other times of the year may inform us of how their distribution changes, and how they move with the currents through the seasons.

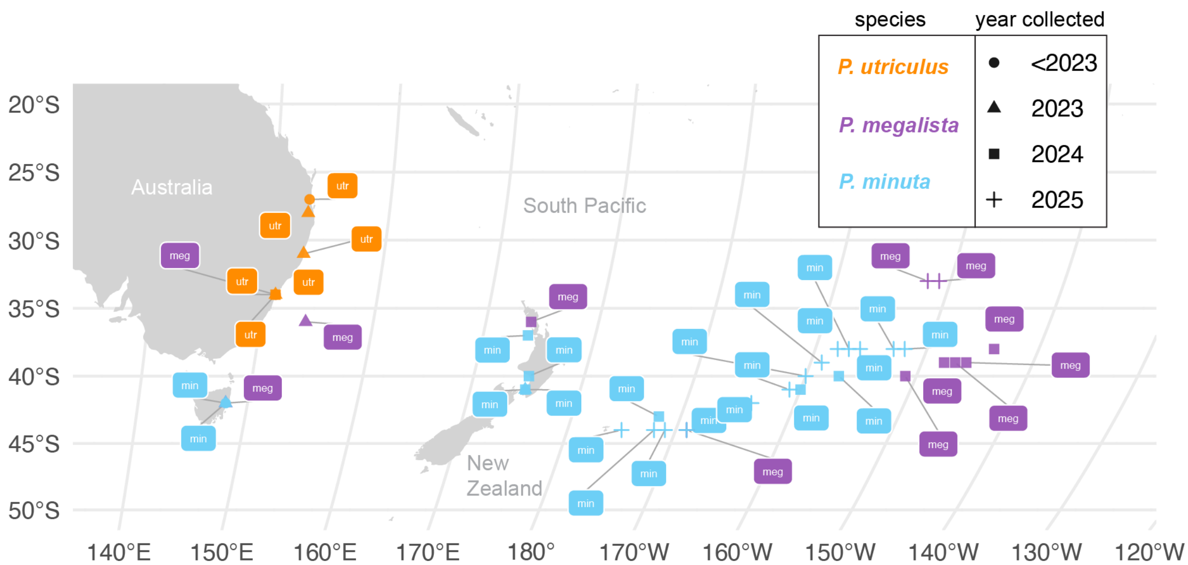

Figure 5: **Open ocean sampling in the S. Pacific shows distinct open water distributions**. Sampling over multiple years shows *P. minuta* in open waters East of New Zealand. In both years (2024 and 2025), observations further East were exclusively of *P. megalista*. meg=*P. megalista*, minuta=*P. minuta*, utr=*P. utriculus*.

Expansion of *P. minuta* range

We also collected live *P. Minuta* specimens from the Southern Atlantic to close gaps in our previous global analysis. We collected specimens from Saint Helena, observed their morphology, and genetically sequenced them. Our results showed that five out of six sequenced samples were *P. minuta*, and one was *P. megalista* (Add a supp figure of a phylo tree). *P. minuta* was previously only observed in the Southwestern Pacific and considered to be a species with a tighter regional distribution. Finding it in the South Atlantic suggests its range is larger than previously understood, at least in rare occurrences. Based on these results, we performed a detailed examination of potentially missed observations of *P. minuta* out of the expected range on iNaturalist. We confirmed isolated stranding events in [**TODO-5: date**] in Chile (SE Pacific), in [**TODO-6: date**] in Western Australia, and [**TODO-7: date**] in Southern Africa. *(#2)* These are consistent with a larger open ocean distribution in the Southern Pacific, Indian, and Atlantic oceans for *P. minuta*.

Morphological characterization of species

We collected, observed, and photographed live (Figure 6 A-H) and fixed (Figure 6 I-R) specimens of each species to establish features for morphological identification. We also propose common names for each species, as the terms "Portuguese man-o'-war" and "bluebottle" are currently used to refer to either species. Based on these results, we present the shared characters and diagnoses of species:

Proposed common names

Physalia have many common names across languages (Pugh 2019), several of which refer to a specific kind of Portuguese sailing ship, the man-o-war or caravala. In the Pacific, especially Australia, *Physalia* are also referred to as bluebottles, evocative of their striking color, and of their 'message-in-a-bottle' floating behavior. The common name Portuguese man-o-war has proven problematic, however, as the reference to the Portuguese ship is easily confused with a comment about their biogeography. In reality, the only *Physalia* species known from Portugal is *P. physalis*, which is likely also the species to which the common name man-o-war was originally attributed. To rectify this situation, and to provide a standard set of English common names that can be used to refer to these species worldwide, we propose the following: for the genus, we propose the common name "bluebottles" for *P. physalis*, we propose this species retains the name "Portuguese man-o-war" for *P. utriculus*, the most widespread species and most abundant in English speaking parts of the Pacific, we propose the name "common bluebottle" for *P. minuta*, we propose the name "little bluebottle" for *P. megalista*, distributed around the southern tips of continents, we propose the name "southern bluebottle" For *P. mikazuki*, several names have already been proposed, including Japanese man-o-war, crescent helmet man-o-war, and crescent moon-o-war (Yongstar et al. 2026). This species is found outside of Japan (including Mexico, Pakistan, and elsewhere), and is also not the only or even most commonly collected Physalia species in Japan. For consistency and clarity, we propose the name "crescent helmet bluebottle" for *P. mikazuki*.

Physalia (bluebottle) characteristics across species

While *Physalia* colonies are often illustrated with the tentacles, palpons, and other zooids draped below the colony ventrally (as seen in the lateral view in Figure 6 A-D), we observe in live specimens another posture common to all species (Figure 6 E-H). In this posture, the posterior zone is flared opposite the main zone (e.g., to the left for right-handed colonies), and the main zone is pulled parallel to the surface of the water. In this arrangement it is apparent that large gonodendra are borne on the ventral side of clusters, such that they hang below the colony (e.g., Figure 6 G), while palpons and tentacles are borne on the dorsal side, such that they insert nearest the surface (e.g., Figure 6 H). When seen from above, the float may have

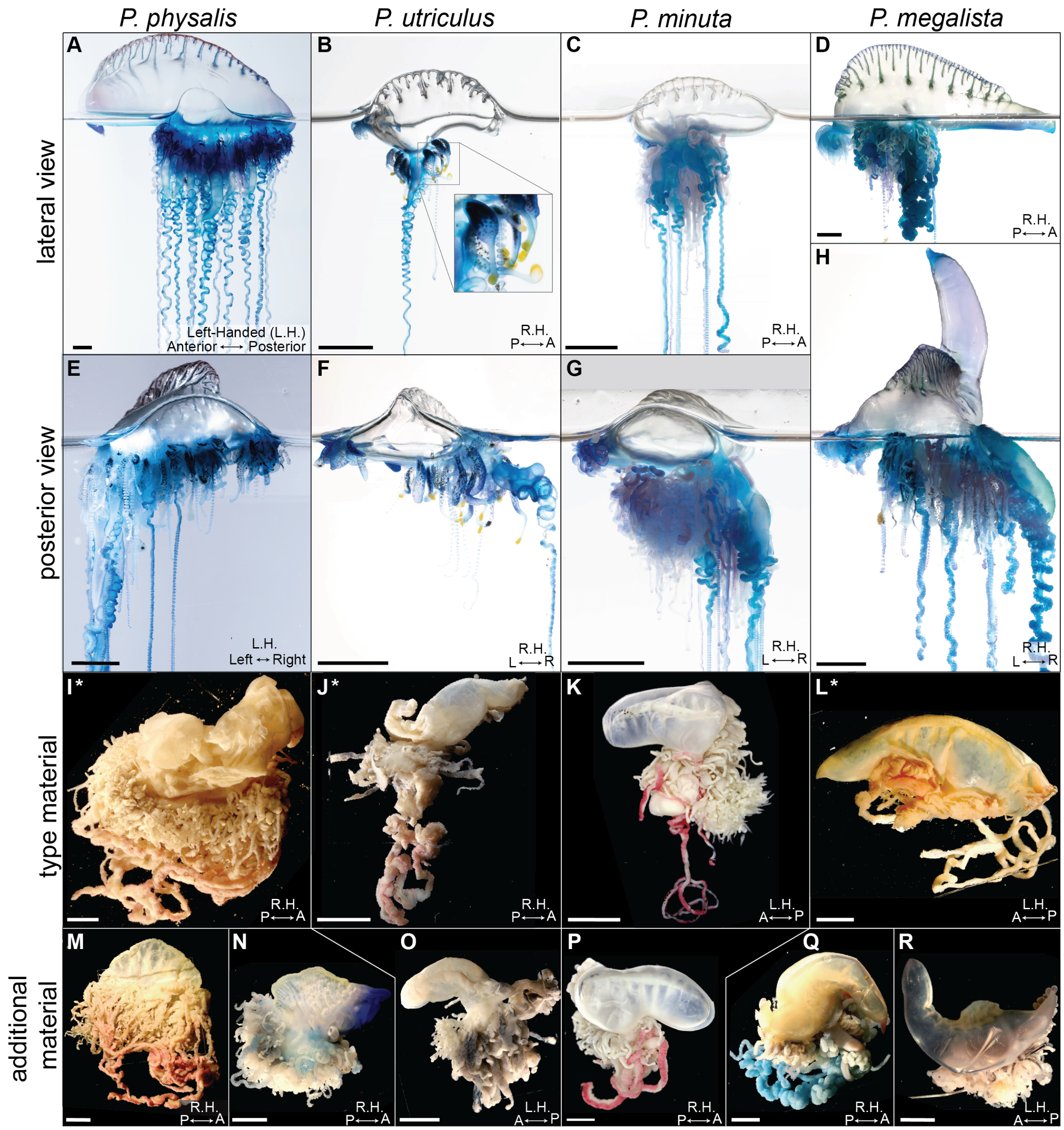

Figure 6: **Comparative morphology of live and fixed specimens of *Physalia*.** (A-D) Lateral views of four Physalia colonies, specimens [**TODO-8: specimen IDs, #2**]. Left-handed specimens have anterior to the left, when viewing the side of the float bearing the main zone of the colony. (E-H) Posterior view, showing a typical floating posture with the posterior and main zones held parallel to the surface, specimens [**TODO-9: specimen IDs, #2**]. (I-L) Type material for each of the four species, viewed laterally, specimens [**TODO-10: specimen IDs, #2**]. Asterisk indicates neotype material designated here. (M-R) Additional specimens examined. Scale bars = 1cm.

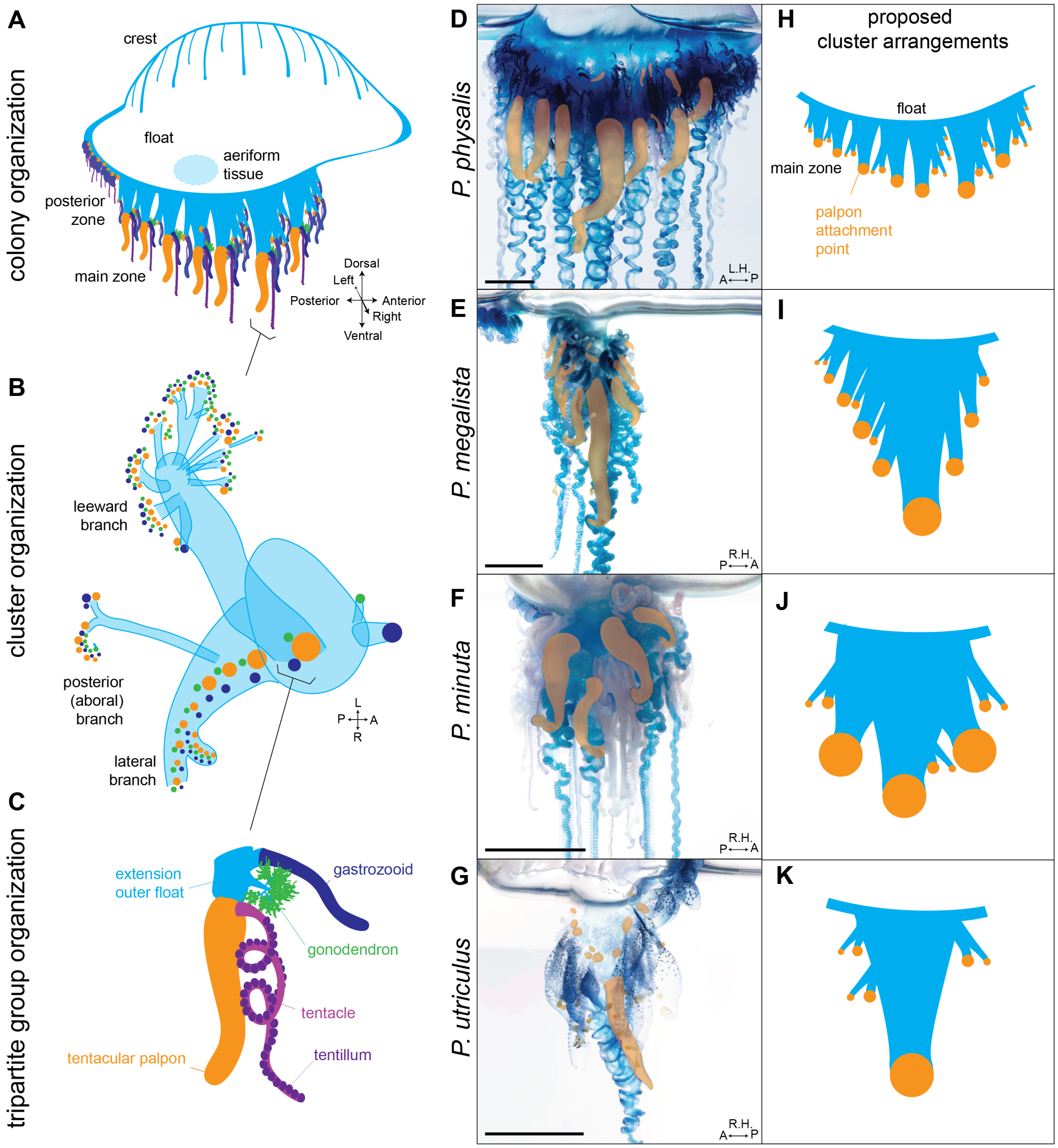

Figure 7: **Colony organization across *Physalia* species.** (A-C) (A) Physalia colonies are made up of a float, crest, and two colony zones (posterior and main), each made up of clusters of zooids. (B) Clusters contain tripartite groups, arranged on branches, diagram modified from (Totton 1960). (C) Tripartite groups contain a gastrozooid, gonodendron, and a tentacle with an associated palpon. (D-G) Images of main zones from each species, all identifiable palpons are false-colored in orange to highlight position. Specimens [**TODO-11: specimen IDs, #2/#7**]. (H-J) Proposed cluster arrangements across species, showing only palpon attachment points to highlight differences in organization.

a sinuous appearance, especially for specimens of *P. utriculus* (**[TODO-12: supp. figures, #23]**).

Historically, one of the challenges to defining morphological differences across specimens was in discerning differences in maturity from differences across populations. This is particularly difficult when considering tentacle numbers. Here we present the following descriptive statements as a paradigm that allows us to address this challenge: [1] As described by Totton (Totton 1960), we consider all *Physalia* colonies to consist of clusters (Totton's "cormidia", see Munro et al for discussion of terminology (Munro et al. 2019)) composed of tripartite groups (Figure 7). Within each cluster, there is a primary tripartite group with branches of additional groups added laterally (Figure 7 B). Each tripartite group consists of a palpon and associated tentacle, a gastrozooid, and a gonodendron (Figure 7 C). [2] While all tentacles are associated with a palpon (Totton 1960), we observe that major tentacles are associated with prominent palpons that can be easily identified, while very small tentacles may have reduced palpons. [3] We define an axis of tentacle maturity that ranges from small tentacles with tentilla spaced apart, appearing as beads on a string (**[TODO: panel]**), to major or major tentacles that have tightly packed tentilla and a coiled appearance.

Using this paradigm, we can define differences in the colony-level organization across species. First, by observing the position of predominant palpons associated with major tentacles, we identified clusters along the main zone of the colony (Figure 7 D-G). Based on the position of these clusters, we observed a primary difference between *P. physalis* and other species is that in the former, clusters are on independent extensions of the outer wall of the float. In *P. megalista*, *P. minuta*, and *P. utriculus*, clusters are typically borne primarily on one extension, though in *P. megalista* this may be divided into two or more, with one larger than the rest. *P. minuta* differs from *P. megalista* and *P. utriculus* in having multiple palpons and associated tentacles of a similar maturity on the extension. In contrast, *P. megalista* and *P. utriculus* show a strong gradient of size extending anterior and posterior to the largest palpon. In typical *P. utriculus* colonies, adjacent palpons and tentacles are much smaller than the central palpon and tentacle (though as mentioned, we observe specimens from Indonesia with more mature adjacent clusters).

Physalia physalis, Portuguese man-o'-war (Figure 6 A, E)

Large *P. physalis* colonies can typically be distinguished from other species by their relatively large crest, running nearly the length of the float, that may be raised to a height as large as the float, and by the main zone of the colony that can have as many as seven clusters of zooids (Totton 1960), each with a principal palpon and associated large tentacle. In a typical floating posture, there is often no visible gap between the main and posterior zone of the colony (Figure 6 E). Clusters are borne on separate extensions of the outer float, which are distributed along the anterior-posterior axis of the body.

Small *P. physalis* may share some characteristics associated with other species, in particular *P. utriculus*. These may reflect shared patterns of growth, for example, small *P. physalis* colonies may also have a single prominent tentacle. Across all species, the float and crest are muscular and can rapidly change shape through contraction, which may lead to postures that are similar to other species. For example, if the anterior end of the float is extended, *P. physalis* may appear to have the anterior projection associated with *P. megalista*. In this paper we have endeavored to describe the typical postures for each species that may reflect underlying differences in tissue-level anatomy.

P. physalis colonies vary in color from blue, to pink, purple, and green, but specimens from the South Atlantic and Caribbean Sea in particular achieve a bright pink coloration that is not known from other species of *Physalia* ([**TODO-13: supp. figure, #23**]). Colonies from both the North and South Atlantic may have reddish gastrozooids, also unique to *P. physalis* ([**TODO-14: supp. figure, #23**]). Other colonies have dark blue, purple, or blue-green gastrozooids and palpons, while gonodendra are light purple. The apex of the crest is often pink, but may have no obvious coloration. The large float is typically shaded blue or pink, and is not glassy or strongly opalescent.

Neotype specimen: YPM-IZ-110254

***Physalia utriculus*, common bluebottle (Figure 6 B, F)**

In our previous publication, *Physalia utriculus* comprised two well-supported sister lineages, labeled B1 and B2 in Church et al. (2025). The neotype presented here for *P. utriculus* (YPM-IZ-110777) is a member of the B1 lineage; B2 was recently described as a separate species, _'P. mikazuki_' by Yongstar et al. (2025) (Yongstar et al. 2025, 2026)

The most reliable positive indicator of specimens of *P. utriculus* is the presence of bright yellow coloration on the distal tip of gastrozooids (Figure 6 B inset). Other than the distal tip, gastrozooids are typically a bright blue, as are palpons and large tentacles, while gonodendra are light purple ([**TODO-15: supp. figure, #23**]). Crest coloration varies from pink to blue or no color ([**TODO-16: supp. figures, #23**]). The anterior apex of the float may be colored green, yellow, or blue ([**TODO-17: supp. figures, #23**]). Excluding the crest and anterior apex, coloration on the float is usually absent, and observation of a glassy transparent float is usually indicative of *P. utriculus* (though very recently surfaced specimens of all species have glassy, globular floats). The float of *P. utriculus* may appear curved or S-shaped when viewed dorsally, though in live animals this can change quickly as the muscles in the float contract and relax.

P. utriculus shares a strong central organization of the main zone with *P. megalista*. All zooids appear to be borne on a single extension of the outer float of the colony, in contrast to *P. physalis* that has multiple extensions. Small individuals of *P. utriculus* may also resemble P. minuta due to glassiness of the float and size, but they do not share the central organization of the main zone.

We propose calling *P. utriculus* a common bluebottle. The term "bluebottle" has been historically used in English-speaking countries on the Indian and Pacific oceans' coasts, where this species has a wide range of distribution. A previous proposed common name for *P. utriculus* was Indo-Pacific bluebottle/man-o'-war, but this species is also found in the Atlantic, making this name inadequate.

Physalia mikazuki, crescent helmet bluebottle ([TODO: panel, pending #7])

The description by Yongstar et al of *Physalia mikazuki* (Yongstar et al. 2025, 2026) specimens (B2 lineage) indicates many shared morphological features with _P. Utriculus_specimens (B1 lineage), such as the distinct bright yellow coloration of the gastrozooid distal tip, bright blue gastrozooids, light purple gonodendra, variable crest and body coloration with a glassy float ([**TODO-18: figure, #23**]). The most prominent characteristic differentiating *P. utriculus* from *P. mikazuki* is the number of major tentacles. *P. utriculus* are often associated with the presence of a single major tentacle, while *P. mikazuki* specimens have multiple. Genetic data have confirmed that rare *P. utriculus* individuals have >1 tentacle (see [**TODO-19: supp. figure, #8**]), and it has been suggested since at least 1910 that tentacle number can vary across colonies of *P. utriculus* (Kawamura 1910; Pugh 2019). Therefore, the tentacle number alone should be treated with caution as a diagnostic character. Here we suggest that a more robust feature that appears to be distinct is the colony arrangement of zooids. In *P. mikazuki*, the main zone of the colony of the specimen shown in the original description has multiple clusters of zooids born from distinct extensions of the float (see comparison below). This is consistent with what we observed in the larger of the two fixed specimens that have been sequenced for genomic analysis [**TODO-20: figure, #7**]

Furthermore, one trait noted as distinct for *P. mikazuki* in the Yongstar description is the number of zooid clusters in the main and posterior zone. However, the number of clusters has not been counted in the *P. utriculus* or other *Physalia* spp. across growth forms or its range, and it is possible this character varies with natural growth, as is true of other features of *Physalia* colonies (Totton 1960). Additionally, the Yongstar description noted a gastrozooid morphology that may be different between P. mikazuki and *P. utriculus* (Yongstar et al. 2025, 2026), however, gastrozooids are highly elastic due to their function and have not been described in other *Physalia* species, making them unreliable for species diagnoses.

Here we report observations of colonies that match the described morphology of *P. mikazuki* , with dense main zones of the colony bearing multiple major tentacles and associated palpons, on colonies with yellow-tipped gastrozooids ([**TODO-21: supp. figure, #23**]). These specimens, observed alive off the southern coast of Indonesia, vary in size (from ~3 to ~10 cm total float length), suggesting this is not correlated with the overall size of the colony. We hypothesize that, if these features prove reliable for *P. mikazuki* identification, the range of this lineage extends throughout the Pacific and Indian Oceans, consistent with preliminary indicators from barcode data (MK084614.1), and our previous work (Church et al. 2025).

Physalia minuta, little bluebottle (Figure 6 C, G)

P. minuta was described in detail in our previous manuscript (Church et al. 2025), but here we add the following observations: The float of beached and fixed specimens often has an elongated cylindrical shape (Figure 6 K, P). In contrast to *P. utriculus*, *P minuta* colonies commonly have multiple principal tentacles and associated palpons at apparently similar levels of maturity, but unlike both *P. utriculus* and *P. mikazuki*, they do not have brightly yellow-tipped gastrozooids. Colony bodies are born from a single extension of the outer wall of the float, such that the main zone of the colony has a "bunched" appearance, as opposed to elongated along either the dorsal/ventral or anterior/posterior axis. The combined characters of multiple large tentacles and palpons, large gonodendra (indicating a mature colony), and a small size (around **[TODO-22: size threshold, #2]** cm) is strongly indicative of *P. minuta*.

Zooid and tentacle coloration varies from bright blue to dark blue-green; gastrozooids may have pale cream-colored distal tips (**[TODO-23: supp. figures, #23]**). The float is typically without color except for the anterior apex, which may be green (**[TODO-24: supp. figures, #23]**). In live specimens, the crest may be raised (Figure 6 C), but in beached or fixed specimens it is typically fully relaxed (Figure 6 P).

Physalia megalista, southern bluebottle (Figure 6 D, H)

P megalista is best recognized by the shape of the float and crest. In typical specimens, the float is elongated, with a prominent projection of the float beyond the anterior point of insertion of the crest (i.e., the crest is "incomplete"). In a relaxed posture, this projection of the float may appear shorter (Figure 6 D), but when disturbed, the muscles are contracted such that the anterior projection is raised perpendicular to the water (Figure 6 H). It is common to find specimens of *P. megalista* in this position on the beach. Another common posture has both the anterior and posterior apices of the flared on the opposite side of the main zone, giving the float a "horned" appearance (**[TODO-25: supp. figures, #23]**). The crest of *P. megalista* is distinct in having a crimped appearance along the dorsal surface (Figure 6 D).

P megalista may have one or multiple prominent tentacles. These are generally born from one extension of the outer float, though in large specimens there may be adjacent smaller extensions (**[TODO-26: supp. figure, #23]**). Like *P. utriculus*, the tentacles and palpons of *P. megalista* exhibit a strong central organization such that the most distal tentacle and palpon are the largest, and those adjacent to the anterior and posterior are progressively smaller. There is also often a prominent gap between the posterior and main zone, as in *P. utriculus* (Figure 6 H). Unlike most *P. utriculus*, however, the main zone of the colony typically has more tentacles and palpons at late stages of maturity (Figure 6 H).

Zooid and tentacle coloration is typically a dark blue-green, with palpons more green than tentacles. In some specimens in the South Atlantic, minor tentacles may have red coloration

([**TODO-27: supp. figure, #23**]). The float is often opalescent purple and blue-green, and rarely glassy. The crest often has a lavender apex with green at the bases of the septae.

Neotype specimen: WAM-Z88480

Discussion

As new *Physalia* species are described from the open ocean (Church et al. 2025; Yongstar et al. 2025, 2026), it becomes critical to provide resources to facilitate their identification and monitoring. To accomplish this purpose, we have provided here an integrated dataset of thousands of classified photographs, detailed descriptions of morphology, curated reference panels of genes and mitochondrial genomes, and maps of the distribution and seasonality of *Physalia* species. We also released an ML model that can be used to diagnose images of beached *Physalia* with a high accuracy (83.8%). Together, this toolkit can be used by citizen scientists, naturalists, and researchers across the globe, including in undersampled regions, such as South and Southeast Asia, Eastern Africa, parts of South America. Our hope is that morphological descriptions and spatiotemporal data will boost interest and accuracy of worldwide iNaturalist community efforts, and advance future efforts to understand the distribution, seasonality, and diversity of *Physalia*.

Our results show that four *Physalia* species (*P. physalis*, *P. utriculus*, *P. minuta*, *P. megalista*) have distinct seasonal patterns and cross-basin distributions. In each case, there are notable species-specific patterns as well as periods of significant overlap in space and time. For example, in the Southwestern Pacific, we observe that *P. utriculus* and *P. minuta* share seasonal patterns of peak abundance in the Southern Hemisphere summer months, but their spatial distribution is quite distinct, with *P. utriculus* nearly absent from New Zealand beaches but highly abundant in Eastern Australia, while *P. minuta* shows the reverse pattern. *P. megalista* in contrast is abundant in both locations and displays a distinct seasonal pattern of abundance. The result is that the dominant species found on a given beach cycles over both space and time. Similar patterns of turnover are observed between *P. megalista* and *P. utriculus* in Southern Africa, and *P. physalis* and *P. utriculus* in the Gulf of Mexico. Our exploration of median day-of-year of observations shows a latitudinal gradient throughout a year for some species (*P. megalista* in Eastern Australia, _P. utriculus in Southern Africa, *P. physalis* in North America). The observed patterns are likely a result of transport processes governed by major currents (e.g. Eastern Australian Current, Agulhas Current, Gulf Stream). More detailed regional surveys across species are needed to understand what drives the observed spatiotemporal patterns. .

Environmental processes that influence distributions of *Physalia* have been the subject of significant research, but their focus is largely on the processes that bring adults to shore where they can have significant economic and health implications(Macías, Prieto, and García-Gorriz 2021; Bourg et al. 2022, 2024; Headlam et al. 2020; Roca et al. 2025; Colaço Martins et al. 2024; Fierro et al. 2021; Carvalho et al. 2026; Hewitt et al. 2026). In adults, movement is mediated by interactions of seasonal wind patterns, gyres, eddies, and currents, but at juvenile stages *Physalia* spend a part of their life cycle under the water surface(Totton 1960). Their reproductive structures detach from the floating animal, then sink to a certain depth where they spawn and fertilize (Richter 1907; Steche 1907; Totton 1960; Oguchi et al. 2024). *Physalia* larvae spend an unknown amount of time in the water column in a pelagic phase until they inflate a float, which raises them to the ocean surface(Totton 1960). Depending on the depth at which the embryo forms, and how fast they emerge to the surface, their movement may be influenced by the subsurface oceanographic processes for a part of their life cycle. To understand how species follow distinct spatiotemporal trends, it will be critical to learn more about their reproductive biology, including finding gametes or larvae at depth to estimate dispersal. Additionally, it may be that seasonal abundance is largely indicative of the presence of preferred prey (generally fish) of each species and life stage. More research on differences in prey specificity and gut contents will shed light on how distinct patterns of seasonality reflect specialization on distinct fish populations.

To advance further studies of different aspects of *Physalia* biology, we also provide comprehensive morphological descriptions of the four species (*P. physalis*, *P. utriculus*, *P. minuta*, and *P. megalista*), and a discussion on recently described *P. mikazuki*. Despite this, several key questions about the morphological differences in *Physalia* still remain unanswered, particularly around the complex lineage of yellow-mouthed species like *P. utriculus* and *P. mikazuki*. Further studies of their reproduction, development, growth patterns, and variation across their distribution will be crucial in establishing how genetic variation tracks morphological variation in coloration, tentacle and zooid number, and overall colony architecture. In particular, we reiterate the caution of previous authors (Totton 1960; Pugh 2019) that, because certain traits vary throughout growth and development, studying the morphology of different life cycle stages across species is essential to understand what fixed differences are between lineages. Without such data, we find it difficult to diagnose specimens of *P. utriculus* from their sister lineages based on the number of tentacles, zooids, or shape of the gastrozooids alone.

Further exploration of the distribution and seasonality in the broad *P. utriculus* lineage in particular has great potential to advance efforts to understand their biology. *P. utriculus* has the broadest distribution of all species, from Texas (USA) to the Pacific coast of Ecuador, and has corresponding patterns of regional diversity in genetic subpopulations (Church et al. 2025). Studying the processes that allow a lineage to remain cohesive across such vast geographic ranges has the potential to shed light on what drives dispersal and gene flow across ocean basins. At the same time, evidence that a sister lineage (*P mikazuki*) with similar morphology overlaps maintains a distinct and identifiable genotype despite strong overlap in ranges suggests that barriers to gene flow can arise in open waters(Church et al. 2025; Yongstar et al. 2025). Further research to distinguish the proximal causes of these barriers can shed light on the origins of the complex and fascinating neustonic community.

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Abedon, River B, Mary Beth Decker, Casey W Dunn, and Samuel H Church. 2025. "From Surfacing to Stranding: The Origins and Dispersal Dynamics of a Neustonic Siphonophore." *bioRxiv*, August.

Bolger, Anthony M, Marc Lohse, and Bjoern Usadel. 2014. "Trimmomatic: A Flexible Trimmer for Illumina Sequence Data." *Bioinformatics* 30 (15): 2114–20.

Bourg, Natacha, Amandine Schaeffer, Paulina Cetina-Heredia, Jasmin C Lawes, and Daniel Lee. 2022. "Driving the Blue Fleet: Temporal Variability and Drivers Behind Bluebottle (Physalia Physalis) Beachings Off Sydney, Australia." *PLoS One* 17 (3): e0265593.

Bourg, Natacha, Amandine Schaeffer, Anne Molcard, Christopher Luneau, Daniel E Hewitt, and Rémi Chemin. 2024. "Ocean Wanderers: A Lab-Based Investigation into the Effect of Wind and Morphology on the Drift of Physalia Spp." *Mar. Pollut. Bull.* 207 (116856): 116856.

Bowen, Brian W, Michelle R Gaither, Joseph D DiBattista, Matthew Iacchei, Kimberly R Andrews, W Stewart Grant, Robert J Toonen, and John C Briggs. 2016. "Comparative Phylogeography of the Ocean Planet." *Proc. Natl. Acad. Sci. U. S. A.* 113 (29): 7962–69.

Carvalho, Patrícia, Clara Cordeiro, Soraia Pereira, and Antonina dos Santos. 2026. "Spatiotemporal Patterns and Environmental Drivers of Physalia Physalis Strandings Along Mainland Portugal (Northeastern Atlantic)." *Estuar. Coast. Shelf Sci.* 329 (109674): 109674.

Casteleyn, Griet, Frederik Leliaert, Thierry Backeljau, Ann-Eline Debeer, Yuichi Kotaki, Lesley Rhodes, Nina Lundholm, Koen Sabbe, and Wim Vyverman. 2010. "Limits to Gene Flow in a Cosmopolitan Marine Planktonic Diatom." *Proc. Natl. Acad. Sci. U. S. A.* 107 (29): 12952–57.

Chan, Patricia P, Brian Y Lin, Allysia J Mak, and Todd M Lowe. 2021. "tRNAscan-SE 2.0: Improved Detection and Functional Classification of Transfer RNA Genes." *Nucleic Acids Res.* 49 (16): 9077–96.

Church, Samuel H, River B Abedon, Namrata Ahuja, Colin J Anthony, Dalila Destanović, Diego A Ramirez, Lourdes M Rojas, et al. 2025. "Population Genomics of a Sailing Siphonophore Reveals Genetic Structure in the Open Ocean." *Curr. Biol.* 35 (15): 3556–3569.e6.

Colaço Martins, Lara, José Nuno Gomes-Pereira, Gisela Dionísio, and Jorge Assis. 2024. "Unravelling Environmental Drivers and Patterns of Portuguese Man o' War (Physalia Physalis) Blooms in Two Ocean Regions: North Atlantic and the Southeast Pacific." *Mar. Pollut. Bull.* 209 (Pt B): 117278.

Damian-Serrano, Alejandro, Elizabeth D Hetherington, C Anela Choy, Steven H D Haddock, Alexandra Lapides, and Casey W Dunn. 2022. "Characterizing the Secret Diets of Siphonophores (Cnidaria: Hydrozoa) Using DNA Metabarcoding." *PLoS One* 17 (5): e0267761.

Donath, Alexander, Frank Jühling, Marwa Al-Arab, Stephan H Bernhart, Franziska Reinhardt, Peter F Stadler, Martin Middendorf, and Matthias Bernt. 2019. "Improved Annotation of Protein-Coding Genes Boundaries in Metazoan Mitochondrial Genomes." *Nucleic Acids Res.* 47 (20): 10543–52.

Dunn, Casey W, and Samuel H Church. 2026. "Sharkmer: Repurposing PCR Primers for Targeted Genome Assembly Using in Silico PCR." *Bioinformatics* 42 (4).

Fierro, Pablo, Loretto Arriagada, Andrea Piñones, and Juan Francisco Araya. 2021. "New Insights into the Abundance and Seasonal Distribution of the Portuguese Man-of-War Physalia Physalis (Cnidaria: Siphonophorae) in the Southeastern Pacific." *Reg. Stud. Mar. Sci.* 41 (101557): 101557.

Gaylord, B, and S D Gaines. 2000. "Temperature or Transport? Range Limits in Marine Species Mediated Solely by Flow." *Am. Nat.* 155 (6): 769–89.

Goetze, Erica. 2003. "Cryptic Speciation on the High Seas; Global Phylogenetics of the Copepod Family Eucalanidae." *Proc. Biol. Sci.* 270 (1531): 2321–31.

———. 2011. "Population Differentiation in the Open Sea: Insights from the Pelagic Copepod Pleuromamma Xiphias." *Integr. Comp. Biol.* 51 (4): 580–97.

Headlam, Jasmine L, Kieran Lyons, Jon Kenny, Eamonn S Lenihan, Declan T G Quigley, William Helps, Michel M Dugon, and Thomas K Doyle. 2020. "Insights on the Origin and Drift Trajectories of Portuguese Man of War (Physalia Physalis) over the Celtic Sea Shelf Area." *Estuar. Coast. Shelf Sci.* 246 (107033): 107033.

Helm, Rebecca R. 2021. "The Mysterious Ecosystem at the Ocean's Surface." *PLoS Biol.* 19 (4): e3001046.

Hewitt, Daniel E, Amandine Schaeffer, Jasmin C Lawes, Sean Kelly, Alistair G B Poore, Moninya Roughan, Saima Ajaz, Timothy C Ingleton, and Kylie A Pitt. 2026. "Blowin' in the Wind: Onshore Winds Drive the Occurrence of Bluebottles (Physalia Spp.) at East Australian Beaches." *Ocean Coast. Manag.* 271 (107931): 107931.

Jin, Jian-Jun, Wen-Bin Yu, Jun-Bo Yang, Yu Song, Claude W dePamphilis, Ting-Shuang Yi, and De-Zhu Li. 2020. "GetOrganelle: A Fast and Versatile Toolkit for Accurate de Novo Assembly of Organelle Genomes." *Genome Biol.* 21 (1): 241.

Johnson, Shannon B, Jacob R Winnikoff, Darrin T Schultz, Lynne M Christianson, Wyatt L Patry, Claudia E Mills, and Steven H D Haddock. 2022. "Speciation of Pelagic Zooplankton: Invisible Boundaries Can Drive Isolation of Oceanic Ctenophores." *Front. Genet.* 13 (October): 970314.

Katoh, Kazutaka, and Daron M Standley. 2013. "MAFFT Multiple Sequence Alignment Software Version 7: Improvements in Performance and Usability." *Mol. Biol. Evol.* 30 (4): 772–80.

Kawamura, T. 1910. ""Bozunira" and "Katsuwo No Eboshi" Rhizophysa and Physalia." *Dobutz Z. Tokyo* 22: 445–54.

Li, Heng. 2013. "Aligning Sequence Reads, Clone Sequences and Assembly Contigs with

BWA-MEM." *arXiv [q-Bio.GN]*, March.

Macías, Diego, Laura Prieto, and Elisa García-Gorriz. 2021. "A Model-Based Management Tool to Predict the Spread of Physalia Physalis in the Mediterranean Sea. Minimizing Risks for Coastal Activities." *Ocean Coast. Manag.* 212 (105810): 105810.

Mills, Scott, David H Lunt, and Africa Gómez. 2007. "Global Isolation by Distance Despite Strong Regional Phylogeography in a Small Metazoan." *BMC Evol. Biol.* 7 (November): 225.

Munro, Catriona, Zer Vue, Richard R Behringer, and Casey W Dunn. 2019. "Morphology and Development of the Portuguese Man of War, Physalia Physalis." *Sci. Rep.* 9 (1): 15522.

Nguyen, Lam-Tung, Heiko A Schmidt, Arndt von Haeseler, and Bui Quang Minh. 2015. "IQ-TREE: A Fast and Effective Stochastic Algorithm for Estimating Maximum-Likelihood Phylogenies." *Mol. Biol. Evol.* 32 (1): 268–74.

Norris, Richard D. 2000. "Pelagic Species Diversity, Biogeography, and Evolution." *Paleobiology* 26 (S4): 236–58.

Nunes, Jorge L S, Diego S Campos, Héllida Negrão Dias, Luiz P N e Silva, Denise M R F Bastos, Tommaso Giarrizzo, Allan Jamesson Silva de Jesus, Jamerson Aguiar-Santos, and Ana Catarina Miranda. 2025. "Predation Strategies of the Portuguese Man-of-War *Physalia Physalis*." *Mar. Ecol.* 46 (6).

Oguchi, Kohei, Gaku Yamamoto, Hisanori Kohtsuka, and Casey W Dunn. 2024. "Physalia Gonodendra Are Not yet Sexually Mature When Released." *Sci. Rep.* 14 (1): 23011.

Peijnenburg, Katja T C A, and Erica Goetze. 2013. "High Evolutionary Potential of Marine Zooplankton." *Ecol. Evol.* 3 (8): 2765–81.

Pugh, P R. 2019. "A History of the Sub-Order Cystonectae (Hydrozoa: Siphonophorae)." *Zootaxa* 4669 (1): zootaxa.4669.1.1.

Purcell, J E. 1984. "Predation on Fish Larvae by Physalia Physalis, the Portuguese Man of War." *Marine Ecology Progress Series* 19: 189–91.

Richter, Walter. 1907. *Die Entwicklung Der Gonophoren Einiger Siphonophoren.* cir.nii.ac.jp.

Roca, J, F Tuya, M Gomez, and F Machín. 2025. "Unravelling Temporality and Environmental Drivers of Jellyfish Presence in an Urban Beach." *J. Sea Res.* 205 (102584): Not Available.

Steche, O. 1907. "Die Genitalanlagen Der Rhizophysalien." *Zeitschrift Für Wissenschaftliche Zoologie* 86 (1): 134–71.

Sunday, Jennifer M, Amanda E Bates, and Nicholas K Dulvy. 2012. "Thermal Tolerance and the Global Redistribution of Animals." *Nat. Clim. Chang.* 2 (9): 686–90.

Totton, A K. 1960. "Studies on' Physalia Physalis'(L.) Pt. 1. Natural History and Morphology.[with Plates.]."

Unal, Ebru, and Ann Bucklin. 2010. "Basin-Scale Population Genetic Structure of the Planktonic Copepod Calanus Finmarchicus in the North Atlantic Ocean." *Prog. Oceanogr.* 87 (1-4): 175–85.

White, Crow, Kimberly A Selkoe, James Watson, David A Siegel, Danielle C Zacherl, and Robert J Toonen. 2010. "Ocean Currents Help Explain Population Genetic Structure." *Proc. Biol. Sci.* 277 (1688): 1685–94.

Wood, Derrick E, Jennifer Lu, and Ben Langmead. 2019. "Improved Metagenomic Analysis with Kraken 2." *Genome Biol.* 20 (1): 257.

Yongstar, Chanikarn, Yoshiki Ochiai, Muhammad Izzat Nugraha, Kei Chloe Tan, Ayane Totsu, Waka Sato-Okoshi, and Cheryl Lewis Ames. 2025. "Physalia Mikazuki Sp. Nov. (Phylum Cnidaria; Class Hydrozoa) Blown into Japan's Northeast (Tohoku) at the Whim of Marine Ecosystem Change." *Front. Mar. Sci.* 12 (1653958).

Yongstar, Chanikarn, Yoshiki Ochiai, Muhammad Izzat Nugraha, Kei Chloe Tan, Ayane Totsu, Waka Sato-Okoshi, and Cheryl Lewis Ames. 2026. "Physalia Mikazuki Sp. Nov. (Phylum Cnidaria; Class Hydrozoa) from Tohoku Japan, and Its Improvised Stomachs." *Journal of Integrated Field Science* 23: 25–29.